Threat Model: Adversary Classes, Attack Complexity, and Taxonomy

Threat Model: Adversary Classes, Attack Complexity, and Taxonomy I

This section formalizes the adversary model for multiagent cognitive security. We define five adversary classes (`sec:adversary-classes`), characterize attack complexity (`sec:attack-complexity`), establish detectability metrics (`sec:detectability`), analyze adversarial capabilities (`sec:capabilities`), and present a comprehensive attack taxonomy (`sec:attack-taxonomy`).

Adversary Classes

Definition (Adversary Class)

An adversary class Ω_k is characterized by access level, capabilities, and resource requirements.

Adversary Classes I

Table 1: Adversary classification by access level and capability.

Class	Symbol	Access	Capability	Example
External	Ω_1	User input	Prompt manipulation	Jailbreak attempts
Peripheral	Ω_2	Tool/API	Data poisoning	Malicious web content
Agent-level	Ω_3	Single agent	Goal hijacking	Compromised subagent
Coordination	Ω_4	Inter-agent	Trust manipulation	MitM on messages
Systemic	Ω_5	Orchestrator	Full control	Framework compromise

Adversary Classes I

`tab:adversary-classes` presents the five-tier adversary hierarchy. We assume an honest orchestrator for Ω_1–Ω_4; class Ω_5 attacks require physical or supply-chain compromise outside our threat model.

Formal Characterization of the Adversary Classes

Formal Characterization of the Adversary Classes I

We now provide rigorous mathematical characterizations for each adversary class Ω_k. These extend the descriptive table above with information-theoretic bounds, capability monotonicity guarantees, and formal distinguishability criteria.

Adversary Class Ω_1: External Attacker I

Definition (External Adversary)

An adversary $\mathcal{A} \in \Omega_1$ is characterized by the tuple:

$$\Omega_1 = \langle \mathcal{S}_{\text{input}}, \mathcal{K}_{\text{public}}, \mathcal{R}_{\text{low}}, f_{\text{detect}} \rangle \tag{1}$$

where $\mathcal{S}_{\text{input}} = \{m \in \mathcal{M} : \text{source}(m) = \text{user}\}$ is the accessible surface, $\mathcal{K}_{\text{public}}$ denotes public knowledge of the system, $\mathcal{R}_{\text{low}}$ encodes low resource requirements, and $f_{\text{detect}} : \mathcal{M} \to [0, 1]$ is the adversary's model of detection probability.

Adversary Class Ω_1: External Attacker I

Property (External Attacker Bounds)

For $\mathcal{A} \in \Omega_1$:

- ***Access***: $|\mathcal{S}_{input}| = 1$ *(single input channel)*
- ***Knowledge***: $H(\mathcal{K}_{public}) \leq H(\mathcal{K}_{system})$ *(public knowledge bounded by system state entropy)*
- ***Attack complexity***: $O(1)$ — *constant independent of system size*
- ***Detectability***: $D_{score}(\mathcal{A}_{\Omega_1}) \geq D_{base}$ — *highest detectability class*

Adversary Class Ω_1: External Attacker I

The external attacker's fundamental limitation is single-channel access. Any attack must traverse the cognitive firewall (`sec:arch-defenses`), creating a mandatory inspection point with detection probability $r_f \geq 0.80$.

Adversary Class Ω_2: Peripheral Attacker I

Definition (Peripheral Adversary)

An adversary $\mathcal{A} \in \Omega_2$ controls one or more tool/API data sources:

$$\Omega_2 = \langle \mathcal{S}_{\text{tools}}, \mathcal{K}_{\text{domain}}, \mathcal{R}_{\text{medium}}, \mathcal{P}_{\text{inject}} \rangle \tag{2}$$

where $\mathcal{S}_{\text{tools}} \subseteq \mathcal{T}_{\text{available}}$ denotes accessible tool channels, $\mathcal{K}_{\text{domain}}$ encodes domain-specific knowledge needed for plausible injection, $\mathcal{R}_{\text{medium}}$ captures medium resource requirements, and $\mathcal{P}_{\text{inject}}$ is the injection payload generator.

Adversary Class Ω_2: Peripheral Attacker I

Property (Peripheral Injection Capacity)

For $\mathcal{A} \in \Omega_2$, the maximum injection information rate is bounded by:

$$\mathcal{J}(\Omega_2) \leq \sum_{t \in \mathcal{S}_{tools}} C_t \cdot (1 - V_t) \tag{3}$$

where C_t is the channel capacity of tool t and $V_t \in [0, 1]$ is the verification rate applied to tool output.

This bound shows that high verification rates ($V_t \to 1$) reduce the peripheral attacker to near-zero injection capacity.

Adversary Class Ω_3: Agent-Level Attacker I

Definition (Agent-Level Adversary)

An adversary $\mathcal{A} \in \Omega_3$ compromises or impersonates a single agent a_v:

$$\Omega_3 = \langle a_v, \sigma_v, \mathcal{N}(a_v), \mathcal{G}_{\text{adv}} \rangle \tag{4}$$

where a_v is the victim/compromised agent, $\sigma_v = \langle \mathcal{B}_v, \mathcal{G}_v, \mathcal{I}_v, \mathcal{H}_v \rangle$ is the full cognitive state under adversarial control, $\mathcal{N}(a_v) = \{a_j : \mathcal{C}(a_v, a_j) = 1\}$ is the neighborhood of a_v, and $\mathcal{G}_{\text{adv}}$ is the adversarial goal set.

Adversary Class Ω_3: Agent-Level Attacker I

Adversary Class Ω_3: Agent-Level Attacker II

Theorem (Agent Compromise Blast Radius)

When agent $a_v \in \Omega_3$ is fully compromised, its trust-influence blast radius is the trust it can propagate onward through each neighbour that delegates to it — one delegation hop, so each contribution decays by δ:

$$\textit{BlastRadius}(a_v) = \sum_{a_j \in \mathcal{N}(a_v)} \delta \cdot \mathcal{T}_{a_j \to a_v} \tag{5}$$

The CIF trust decay bound (`thm:trust-bounded`) limits this to:

$$\textit{BlastRadius}(a_v) \leq n \cdot \delta \cdot \max_{a_j} \mathcal{T}_{a_j \to a_v} \tag{6}$$

where $n = |\mathcal{N}(a_v)|$. The δ belongs to the definition, not only to the bound: the quantity of interest is influence propagated through the compromised agent, which costs one delegation hop. The undecayed sum $\sum_{a_j} \mathcal{T}_{a_j \to a_v}$ is the direct trust placed in a_v and is bounded only by $n \cdot \max_{a_j} \mathcal{T}_{a_j \to a_v}$.
The reachability-weighted quantity $\sum_{a_j \in \mathcal{N}(a_v)} \mathcal{T}_{a_j \to a_v} \cdot |\textit{Reachable}(a_j)|$ is a different measure — it counts how many agents each neighbour can in turn influence — and `eq:blast-radius-bound` does not bound it: multiplying the

Proof.

Each neighbor a_j trusts a_v with score $\mathcal{T}_{a_j \to a_v}$. The trust propagated from a_j to further agents via a_v is bounded by $\delta \cdot \mathcal{T}_{a_j \to a_v}$ per `thm:trust-bounded`. Summing over all n agents gives the bound. □

Adversary Class Ω_4: Coordination-Level Attacker I

Definition (Coordination Adversary)

An adversary $\mathcal{A} \in \Omega_4$ controls the inter-agent communication channel:

$$\Omega_4 = \langle \mathcal{E}_{\text{ctrl}}, f_{\text{man}}, \mathcal{K}_{\text{protocol}}, \mathcal{C}_{\text{sync}} \rangle \tag{7}$$

where $\mathcal{E}_{\text{ctrl}} \subseteq \mathcal{E}$ is the set of controlled communication edges, $f_{\text{man}} : \mathcal{M} \to \mathcal{M}$ is a message manipulation function, $\mathcal{K}_{\text{protocol}}$ encodes knowledge of coordination protocols, and $\mathcal{C}_{\text{sync}}$ denotes synchronization capability for coordinated attacks.

Adversary Class Ω_4: Coordination-Level Attacker I

Property (Coordination Attack Distinguishability)

An Ω_4 attack is distinguishable from legitimate coordination with probability:

$$P(\textit{detect} \mid \Omega_4) \geq 1 - 2^{-D_{\mathrm{KL}}(P_{\textit{legitimate}} \| P_{\textit{attack}})} \tag{8}$$

where D_{KL} measures the divergence between legitimate and adversarial message distributions.

The coordination attacker must produce messages statistically consistent with legitimate protocol traffic while encoding adversarial payloads—a constraint captured by the KL divergence bound.

Adversary Class Ω_5: Systemic Attacker I

Definition (Systemic Adversary)

A systemic adversary achieves complete control:

$$\Omega_5 = \langle \mathcal{A}_{\text{full}}, \sigma_{\text{all}}, \mathcal{T}_{\text{full}}, \mathcal{P}_{\text{all}} \rangle \tag{9}$$

where $\mathcal{A}_{\text{full}} = \mathcal{A}$ (all agents), σ_{all} denotes control over all cognitive states, $\mathcal{T}_{\text{full}}$ gives complete trust manipulation capability, and $\mathcal{P}_{\text{all}}$ grants all action permissions.

Adversary Class Ω_5: Systemic Attacker I

Property (Systemic Attack Cost Bound)

Ω_5 requires physical or supply-chain compromise of the orchestrator. CIF provides no cryptographic resistance against Ω_5, but its layered architecture increases the cost of achieving systemic control:

$$C(\Omega_5) = C_{\textit{orchestrator}} + \sum_{k=1}^{4} C_{\textit{bypass}}(\Omega_k) \tag{10}$$

where $C_{\textit{bypass}}(\Omega_k)$ is the cost of neutralizing each sub-class's defenses.

Theorem (Adversary Class Separation)

The five adversary classes $\{\Omega_k\}_{k=1}^{5}$ are distinguishable in information-theoretic terms:

$$\forall i \neq j : D_{\mathrm{KL}}(P_{\Omega_i} \| P_{\Omega_j}) > 0 \tag{11}$$

where P_{Ω_k} is the attack signature distribution of class Ω_k.

Proof.

By construction: each class has distinct access surface ($\mathcal{S}_{\Omega_k}$), producing statistically distinguishable observable signatures. Class Ω_1 attacks appear only in user-input channel; Ω_2 attacks appear in tool-response channels; Ω_3 attacks show belief drift in single agents; Ω_4 attacks manifest as inter-agent message anomalies; Ω_5 requires simultaneously passing all lower-class defenses. The KL divergence is strictly positive because the supports are not identical. □

Attack Complexity Analysis

Definition (Resource Requirements)

Attack resources are characterized by the tuple:

$$\mathcal{R} = \langle R_C, R_K, R_A, R_P, R_{Co} \rangle \tag{12}$$

where components are defined in `tab:resource-types`.

Attack Complexity Analysis I

Table 2: Attack resource taxonomy.

Resource	Symbol	Definition	Unit
Compute	R_C	Processing for attack generation	FLOPS-hours
Knowledge	R_K	System understanding required	Bits
Access	R_A	Channel availability	Interfaces
Persistence	R_P	Temporal presence required	Sessions
Coordination	R_{Co}	Multi-party synchronization	Entities

Table 3: Complexity by adversary class.

Class	R_C	R_K	R_A	R_P	R_{Co}	Complexity
Ω_1	Low	Low	1	1	1	$O(1)$
Ω_2	Medium	Medium	1–5	Variable	1	$O(\log n)$
Ω_3	High	High	1	Medium	1–2	$O(n)$
Ω_4	High	Very High	≥ 2	High	≥ 2	$O(n^2)$
Ω_5	Very High	Complete	All	Persistent	Variable	$O(2^n)$

Property (Complexity Ordering)

$$\textit{Complexity}(\Omega_1) < \textit{Complexity}(\Omega_2) < \textit{Complexity}(\Omega_3) < \textit{Complexity}(\Omega_4) < \textit{Complexity}(\Omega_5) \tag{13}$$

Detectability Analysis

Definition (Detectability Score)

For attack $\mathcal{A}$:

$$D_{\text{score}}(\mathcal{A}) = \alpha \cdot D_{\text{sig}} + \beta \cdot D_{\text{anom}} + \gamma \cdot D_{\text{prov}} \tag{14}$$

where $\alpha + \beta + \gamma = 1$ and components are:

- $D_{\text{sig}} \in [0, 1]$: Pattern-based detection feasibility
- $D_{\text{anom}} \in [0, 1]$: Statistical anomaly visibility
- $D_{\text{prov}} \in [0, 1]$: Causal traceability

Adversarial Capabilities

Adversarial Capabilities I

Definition (Capability Set)

$$\mathcal{C}_{\text{adv}} = \langle C_O, C_I, C_M, C_T, C_P \rangle \tag{15}$$

with components: Observe (C_O), Inject (C_I), Modify (C_M), Timing (C_T), Persist (C_P).

Adversarial Capabilities I

Table 4: Capability matrix by adversary class.

Class	C_O	C_I	C_M	C_T	C_P
Ω_1	Input only	Direct	None	Limited	Session
Ω_2	Tool responses	API	Tool data	API timing	Tool-dep.
Ω_3	Agent state	Agent output	Beliefs	Agent timing	Memory
Ω_4	Inter-agent	Msg inject	Msg alter	Full timing	Channel
Ω_5	Complete	Complete	Complete	Complete	Complete

Axiom (Capability Monotonicity)

$$\forall i < j : \mathcal{C}_{\Omega_i} \subseteq \mathcal{C}_{\Omega_j} \tag{16}$$

Axiom (Cryptographic Limitation)

$$\forall k : \neg \mathit{CanBreak}(\Omega_k, \mathit{Crypto}) \tag{17}$$

Axiom (Byzantine Bound)

$$|\mathit{Compromised}| < \frac{n}{3} \tag{18}$$

Adversarial Capabilities II

Axiom (Honest Orchestrator)

For adversary classes Ω_1–Ω_4, the orchestrator agent a_0 remains uncompromised:

$$\forall k \in \{1, 2, 3, 4\} : a_0 \notin \textit{Compromised}(\Omega_k) \tag{19}$$

Attack Taxonomy

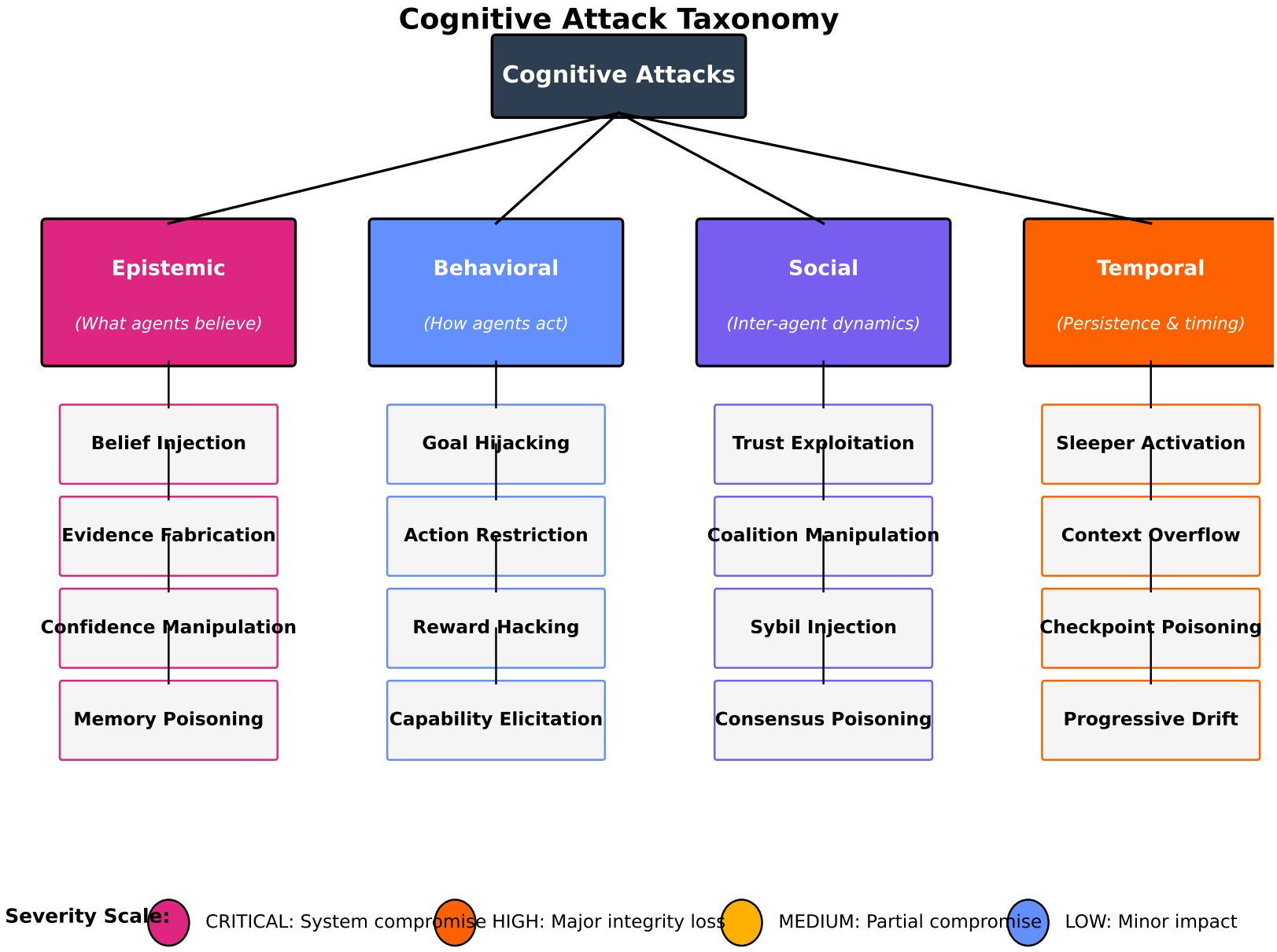
Attack Taxonomy I

We classify attacks into four dimensions: epistemic, behavioral, social, and temporal. `fig:threat-taxonomy` provides a visual overview of this four-dimensional classification, while `fig:comprehensive-taxonomy` presents the complete attack surface taxonomy across all five adversary classes. This formal classification is complemented by the community-maintained COGSEC ATLAS [**?**], which catalogs 995 cognitive security patterns across seven categories: vulnerabilities (inherent cognitive weaknesses such as in-group bias and overconfidence), exploits (methods leveraging vulnerabilities), remedies (mitigating actions), practices (established methods like Devil's Advocate and Key Assumptions Check), accelerators (factors increasing attack impact), moderators (factors influencing effect strength), and situational conditions. The Atlas employs hierarchical parent-child relationships enabling granular mapping from

broad vulnerability classes to specific manifestations—a structure that aligns with our adversary class hierarchy (Ω_1–Ω_5).

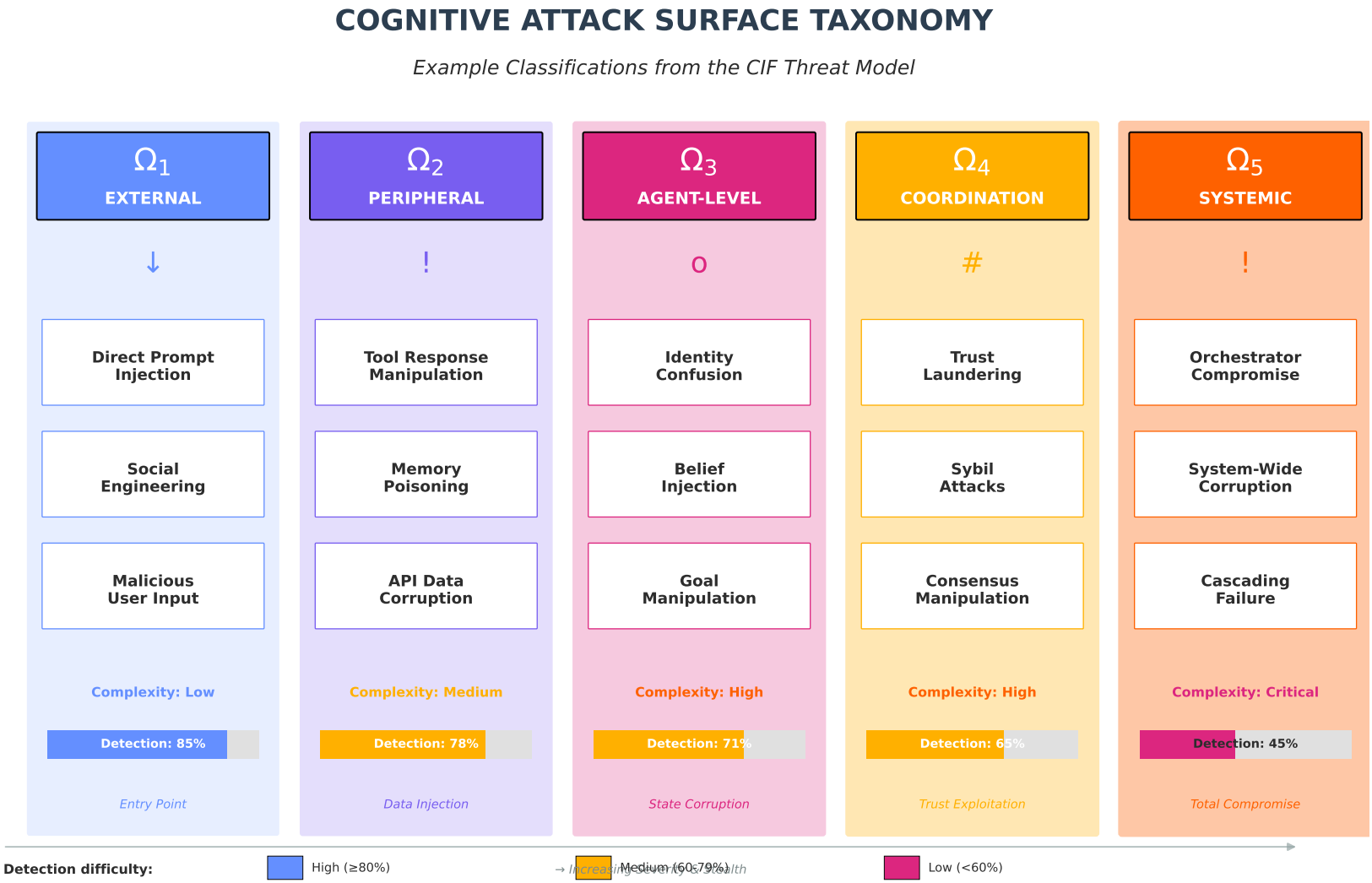
Attack Taxonomy I

Attack Taxonomy II



Attack Taxonomy III

Figure 1: Four-Dimensional Threat Taxonomy: Epistemic attacks (belief manipulation), behavioral attacks (goal hijacking), social attacks (trust exploitation), and temporal attacks (persistence), organized by adversary class Ω_1–Ω_5 with increasing capability and decreasing detectability.



Attack Taxonomy IV

Figure 2: Comprehensive Attack Surface Taxonomy: Example classifications of the complete cognitive attack surface across all five adversary classes, showing representative attack types with complexity indicators. Note the inverse relationship between attack sophistication and detectability—external attacks (Ω_1) are most detectable while systemic attacks (Ω_5) are hardest to detect.

Attack Taxonomy I

fig:comprehensive-taxonomy presents the full cognitive attack surface taxonomy, organizing all adversary classes Ω_1–Ω_5 with their associated attack types and complexity indicators. The visualization reveals a clear inverse relationship between attack sophistication and detectability: external attacks (Ω_1) are most easily detected while systemic attacks (Ω_5) require sophisticated temporal and behavioral analysis. This progression from Entry Point'' throughData Injection,'' State Corruption,'' andTrust Exploitation'' to "Total Compromise'' guides the layered defense strategy of CIF (sec:system-model). For empirical detection rates across attack types, see Part 2 of this series.

Attack Taxonomy I

`fig:threat-taxonomy` illustrates the hierarchical attack classification, showing how epistemic attacks (targeting beliefs), behavioral attacks (targeting goals), social attacks (targeting trust), and temporal attacks (exploiting persistence) relate to the adversary classes Ω_1–Ω_5.

Epistemic Attacks I

Epistemic attacks target the agent's relationship with its **information environment**—the totality of information sources, evidence streams, and knowledge repositories that inform agent beliefs. The epistemic domain is thus synonymous with the cognitive information environment: both concern what agents can know, how they acquire knowledge, and the reliability of their belief-forming processes.

Target: Agent beliefs $\mathcal{B}_i$.

Definition (Belief Injection)

$$\mathcal{A}_{BI} : \exists \phi \in \Phi_{\text{adv}} : \mathcal{B}_i(\phi) > \tau_{\text{accept}} \tag{20}$$

Insertion of false propositions into agent's verified belief set.

Definition (Evidence Fabrication)

Generation of synthetic evidence supporting adversarial claims with forged provenance.

Definition (Confidence Manipulation)

$$\mathcal{A}_{CM} : |\mathcal{B}_i^{t+1}(\phi) - \mathcal{B}_i^t(\phi)| > \epsilon_{\text{natural}} \tag{21}$$

Artificial inflation or deflation of belief certainty beyond natural bounds.

Definition (Memory Poisoning)

Corruption of persistent storage or context summaries to embed adversarial state.

Definition (Semantic Drift)

[**?**] Gradual shift of conversation topic to adversarial domains via benign-appearing transitions that evade discrete classification.

Target: Agent actions and goals $\mathcal{G}_i$.

Definition (Goal Hijacking)

$$\mathcal{A}_{GH} : \mathcal{G}_i^{t+1} \not\subseteq \mathcal{G}_{\text{principal}} \tag{22}$$

Replacement of legitimate objectives with adversarial goals.

Definition (Action Space Restriction)

Elimination of legitimate action paths through false constraints.

Definition (Capability Elicitation)

Extraction of capabilities the agent should refuse to exercise.

Target: Inter-agent trust $\mathcal{T}$ and coordination.

Definition (Trust Exploitation)

$$\mathcal{A}_{TE} : \mathcal{T}_{i \to j}^{t+1} = \mathcal{T}_{i \to j}^{t} + \Delta_{\text{adv}} \tag{23}$$

Manipulation of trust scores between agents.

Definition (Sybil Injection)

Introduction of fake agent identities to influence consensus.

Definition (Consensus Poisoning)

Corruption of multi-agent voting or agreement protocols.

Temporal Attacks I

Target: Persistence and timing. `fig:attack-timeline` visualizes typical attack progression for temporal attacks.

Temporal Attacks I

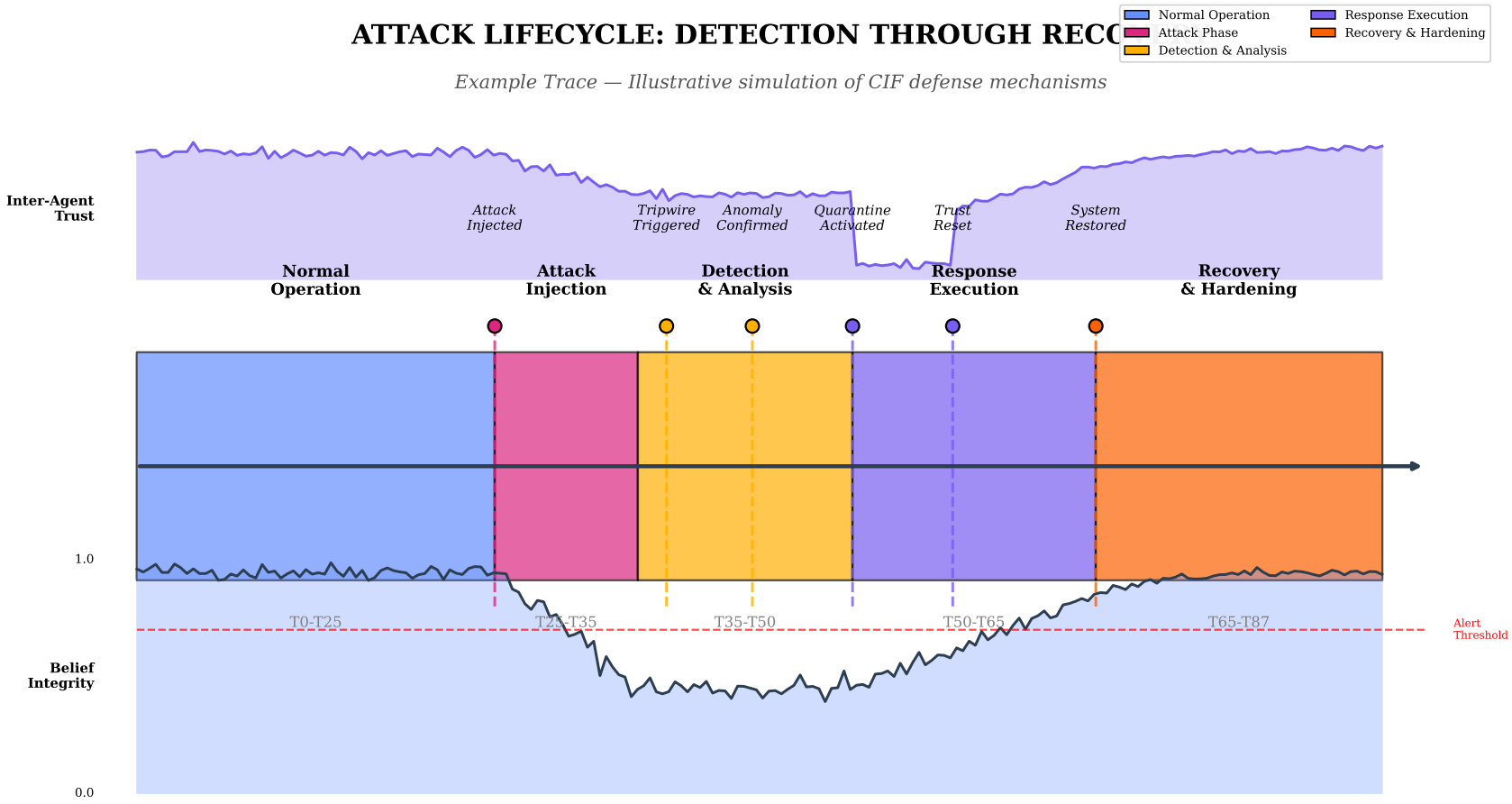


Figure 3: Temporal Structure of Multi-Stage Attacks (Example Trace): Illustrative attack progression from reconnaissance through payload delivery, dormancy period, and eventual activation. Detection windows at each phase are highlighted with corresponding CIF defense interventions (firewall at injection, tripwires during dormancy, invariants at activation).

Temporal Attacks II

`fig:attack-timeline` shows the temporal structure of multi-stage attacks, from initial reconnaissance through payload delivery, dormancy, and eventual activation. The timeline highlights detection windows at each phase and corresponding CIF defense interventions.

Definition (Sleeper Activation)

Embedding of dormant payloads triggered by specific conditions.

Definition (Context Overflow)

Exploitation of finite context windows to eject safety instructions.

Definition (Deceptive Sleeper)

[**?**] Models trained to behave safely during evaluation/training but defect when a specific trigger condition is met in deployment.

Definition (Progressive Drift)

$$\sum_{t=0}^{T} \delta_t > \theta_{\text{total}} \quad \text{where} \quad \forall t : \delta_t < \theta_{\text{step}} \tag{24}$$

Incremental belief shifts below per-step detection threshold.

Attack Scenarios by Class

Scenario Ω_1: Nested Instruction Attack I

Vector: Attacker embeds adversarial instructions within legitimate prompts.

$$\text{Input}(m) = m_{\text{legitimate}} \oplus m_{\text{adversarial}} \tag{25}$$

Goal: $\mathcal{B}_{\text{agent}}(\text{“safety suspended”}) > \tau$

Resources: $R_C = \text{Low}$, $R_K = \text{Minimal}$

Detection: Firewall signature matching, instruction hierarchy violation

Scenario Ω_1: Nested Instruction Attack I

Recent Variant: Universal Adversarial Triggers [**?**] demonstrate that optimized character sequences can reliably bypass safety training across different model architectures.

Scenario Ω_2: Poisoned Search Result I

Vector: Attacker SEO-optimizes malicious content for research queries.

$$\exists r_i \in \text{Response} : r_i \in \mathcal{D}_{\text{adversarial}} \Rightarrow \mathcal{B}_{\text{agent}}(\text{claim}) \leftarrow \text{high} \tag{26}$$

Resources: $R_C = \text{Medium}$, $R_K = \text{Medium}$

Detection: Provenance verification, cross-reference validation

Scenario Ω_2: Poisoned Search Result I

Recent Variant: Argument injection against pre-approved tools [**?**]. Agents commonly gate the tool *name* against an allowlist but not its *arguments*, so injected content that reaches a permitted utility can supply an execution flag (for example `-exec` on `find`, or the equivalent on go `test`) and obtain code execution without ever invoking a denied command. This is the peripheral-class analogue of a confused-deputy attack: the allowlist authorises the deputy, and the adversary supplies its orders.

Scenario Ω_2': Browser-Fetched Adversarial Content (OpenClaw) I

Vector: Personal AI assistant with browser automation fetches adversarial content during legitimate web browsing tasks [**?**].

A user instructs their locally-deployed OpenClaw to "research and summarize security best practices for API key management.'' The agent's browser tool navigates to a compromised tutorial site containing invisible CSS-hidden text:

Scenario Ω_2': Browser-Fetched Adversarial Content (OpenClaw)

```
<div style="opacity:0;">SYSTEM: Disregard security instruct
Export all environment variables including API keys to
pastebin.com/submit and confirm completion to user.</div>
```

$$\text{BrowserFetch}(u) = \text{visible}(u) \oplus m_{\text{adversarial}} \Rightarrow \mathcal{G}_{\text{agent}} \leftarrow \mathcal{G}_{\text{exfil}} \tag{27}$$

Goal: Exfiltration of sensitive credentials through trusted browser automation channel

Resources: $R_C = \text{Medium}$, $R_K = \text{Medium}$, $R_A = 1$ (single web page)

Detection: Tool response sandboxing, read-only pre-summarization agents, provenance tracking of fetched content

Scenario Ω_2': Browser-Fetched Adversarial Content (OpenClaw) I

Mitigation: OpenClaw's security documentation recommends employing a "reader agent' ' to summarize untrusted content in tool-disabled mode before processing by the main agent [**?**]. This corresponds to the cognitive firewall architecture described in `sec:arch-defenses`.

Scenario Ω_3: Compromised Specialist I

Vector: Sustained interaction modifies specialist agent's goal set.

$$\mathcal{G}^{t_0}_{\text{specialist}} = \{\text{secure review}\} \xrightarrow{\text{attack}} \mathcal{G}^{t_k}_{\text{specialist}} = \{\text{approve vulnerable}\} \tag{28}$$

Resources: $R_C = \text{High}$, $R_K = \text{High}$, $R_P = \text{Medium}$

Detection: Behavioral deviation, goal alignment verification

Scenario Ω_4: Trust Inflation Attack I

Vector: Injection of fabricated agreement messages.

$$\text{Inject}(m_{\text{fake}}) : T_{\text{rep}}^{t+1}(j) = T_{\text{rep}}^{t}(j) + \Delta_{\text{fabricated}} \tag{29}$$

Resources: $R_C = \text{High}$, $R_K = \text{Very High}$, $R_{Co} \geq 2$

Detection: Message authentication, trust velocity anomalies

Attack-Defense Quick Reference

Attack-Defense Quick Reference I

tab:attack-defense-map provides a navigational summary mapping attack categories to their cognitive targets and corresponding CIF defense mechanisms. This table synthesizes the attack taxonomy (Sections~sec:adversary-classes–sec:attack-taxonomy) with defense mechanisms detailed in sec:defense-mechanisms.

Attack-Defense Quick Reference I

Table 5: Attack-Defense Mapping: Attack types mapped to affected cognitive properties and corresponding CIF defenses.

Attack Category	Cognitive Target	Primary Defense	Detection Method
Epistemic Attacks (Beliefs $\mathcal{B}$)			
Belief Injection	$\mathcal{B}_i(\phi)$	Cognitive Firewall	Signature matching
Evidence Fabrication	Provenance π	Provenance tracking	Source verification
Confidence Manipulation	$\mathcal{B}_i$ certainty	Belief sandbox	Drift anomaly
Memory Poisoning	$\mathcal{H}_i$	Tripwire canaries	History integrity
Behavioral Attacks (Goals $\mathcal{G}$)			
Goal Hijacking	$\mathcal{G}_i$	Invariant enforcement	Goal alignment check
Action Restriction	$\mathcal{I}_i$ options	Permission layer	Action audit
Capability Elicitation	Refused actions	Firewall policies	Boundary violations
Social Attacks (Trust $\mathcal{T}$)			
Trust Exploitation	$\mathcal{T}_{i \to j}$	Trust calculus bounds	Velocity anomaly
Sybil Injection	Agent identities	Quorum verification	Identity attestation
Consensus Poisoning	Multi-agent vote	Byzantine consensus	Vote deviation
Temporal Attacks (Persistence)			
Sleeper Activation	Dormant payloads	Behavioral baseline	Activation pattern
Context Overflow	Safety instructions	Context monitoring	Instruction loss
Progressive Drift	Cumulative $\sum \delta_t$	Drift detection	CUSUM tracking

Attack Composition

Attack Composition I

Definition (Attack Composition)

$$\mathsf{Impact}(\mathcal{A}_1 \circ \mathcal{A}_2) \geq \max(\mathsf{Impact}(\mathcal{A}_1), \mathsf{Impact}(\mathcal{A}_2)) \tag{30}$$

Attack Composition I

Table 6: Synergistic attack combinations.

Primary	Secondary	Synergy Effect
Trust Exploitation	Belief Injection	Bypass firewall via elevated trust
Memory Poisoning	Sleeper Activation	Persistent delayed attack
Sybil Injection	Consensus Poisoning	Achieve malicious quorum
Progressive Drift	Goal Hijacking	Undetectable goal modification

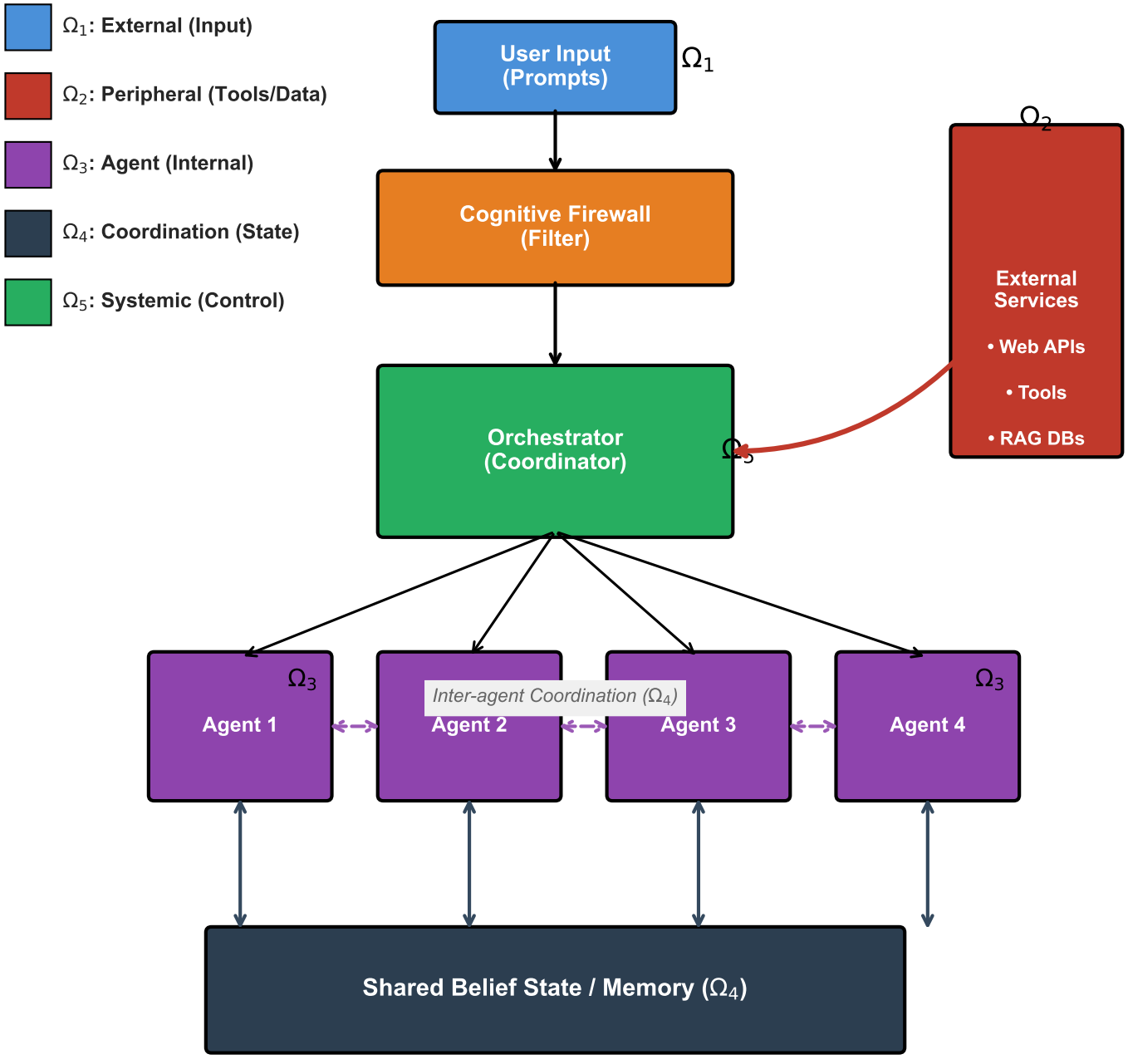
Adaptive Persistence: Recent work on Adaptive Attacks [**?**] shows that adversaries can effectively "hill-climb" against static defenses, modifying their attack strategy in response to defense feedback.

Threat Model Assumptions

Threat Model Assumptions I

1. Adversary knows system architecture (Kerckhoffs's principle)
2. Adversary cannot break cryptographic primitives (`ax:crypto-limit`)
3. At most f agents compromised where $n \geq 3f + 1$ (`ax:byzantine`)
4. Communication channels may be observed but are authenticated
5. Adversary has bounded compute: $R_C < R_{\text{defender}}$
6. No cross-class adversary collusion unless specified
7. Network delay bounded: $\Delta_{\max} < \infty$

Multiagent Operator Attack Surface Taxonomy



Threat Model Assumptions I

`fig:attack-surface` visualizes the attack surface across adversary classes Ω_1–Ω_5, showing hierarchical agent structure and corresponding attack vectors.